

Q: 2(i)(or)
A: 2(i)(or)

(a) $(-27)^{\frac{1}{3}}$

$(-3^3)^{\frac{1}{3}}$

-3 or $\frac{1}{3}$

(c) $(\sqrt[3]{-8})^9$

$= ((-8)^{\frac{1}{3}})^9$

$= (((-2)^{\frac{1}{3}})^3)^9$

$= (-2)^9$

$= -512$

(b) $(\sqrt{16})^4$

$(4)^4$

256

Q: 2(ii)
A: 2(ii)

(a) $4 \times \frac{1}{4} = 1$

$1 \times \frac{1}{1} = 1$

$1 = 1$ (multiplicative inverse identity)

(b) $6 + 0 = 6$

$6 = 6$ (additive property)

(c) $2 \times (3 \times 5) = (2 \times 3) \times 5$

$2 \times (15) = (6) \times 5$

$30 = 30$

Associative property
with respect to multiplication
Property

Q: 2 (iii) _____
 A: 2 (iii) _____

$$(a) \log \frac{5pq^2}{xy^3}$$

$$\log 5 + \log p + \log q^2 - \log x - \log y^3$$

$$= \log 5 + \log p + 2 \log q - \log x - 3 \log y$$

$$(b) 3 \log x - 5 \log y$$

$$\log x^3 - \log y^5$$

$$\log \frac{x^3}{y^5}$$

Q: 2 (iv) (or) _____
 A: 2 (iv) (or) _____

$$\log_b 0.024$$

$$\log_b \frac{24}{1000}$$

$$\log \frac{2^3 \times 3}{10^3}$$

$$\frac{\log 2^3 \times 3}{5^3 \times 2^3}$$

$$\cancel{\log 2^3} + \cancel{\log 3} \quad \log \frac{3}{5^3}$$

$$\begin{aligned} \log 3 - \log 5^3 \\ \log 3 - 3 \log 5 \\ 0.4771 - 3(0.6990) \\ 0.4771 - 2.097 \\ -1.6199 \end{aligned}$$

$$\begin{aligned} \text{Q: } 2(\sqrt{2}) \text{ (or)} \\ \text{A: } 2(\sqrt{2}) \text{ or } \end{aligned}$$

$$(y^2 + 2y - 3)(y^2 + 2y + 11) + 48$$

$$\begin{aligned} \text{Let } y^2 + 2y &= a \\ (a-3)(a+11) + 48 \\ a(a+11) - 3(a+11) + 48 \\ a^2 + 11a - 3a - 33 + 48 \\ a^2 + 8a + 15 \\ a^2 + 3a + 5a + 15 \\ a(a+3) + 5(a+3) \\ (a+5)(a+3) \end{aligned}$$

Putting back values:-

$$(y^2 + 2y + 5)(y^2 + 2y + 3)$$

$$\text{Q. 2 (vi) (or)} \\ \text{A. 2 (vi) (or)} \\ 8 + 12t + 6t^2 + t^3$$

$$t^3 + 6t^2 + 12t + 8$$

perfect cube:- $(t+2)^3 = (t+2)(t+2)(t+2)$

$$(t)^3 + 3(t^2)(2) + 3(t)(4) + (2)^3$$

$$t^3 + 6t^2 + 12t + 8$$

so $8 + 12t + 6t^2 + t^3 = (t+2)^3$
 $= (t+2)(t+2)(t+2)$

$$\text{Q. 2 (vii)} \\ \text{A. 2 (vii)}$$

$$(2x^2 - 3x + 6)(2x^2 - 3x) - 55$$

let $2x^2 - 3x = a$

$$(a+6)(a) - 55$$

$$a(a+6) - 55$$

$$a^2 + 6a - 55$$

$$a^2 + 11a - 5a - 55$$

$$a(a+11) - 5(a+11)$$

$$(a+11)(a-5)$$

$$(2x^2 - 3x + 11)(2x^2 - 3x - 5)$$

$$(2x^2 - 3x + 11)(2x^2 - 5x + 2x - 5)$$

$$(2x^2 - 3x + 11)[x(2x - 5) + 1(x - 5)]$$

H.C.F.:- $(2x^2 + 3x + 11)(x+1)(2x-5)$

Q.2 (viii) (or)
A.2 (viii) (or)

$$5 - |5y + 1| = -9$$

$$-|5y + 1| = -9 \Rightarrow 5$$

$$|5y + 1| = 14$$

$$\begin{aligned} 5y + 1 &= 14 & 2 \quad 5y + 1 &= -14 \\ 5y &= 14 - 1 & 5y &= -14 - 1 \\ 5y &= 13 & 5y &= -15 \\ y &= \frac{13}{5} & y &= \frac{-15}{5} \end{aligned}$$

$$\left\{ \frac{13}{5}, -3 \right\}$$

Q.2 (ix) (or)
A.2 (ix) (or)

$$4(2y + 3) - (6y - 1) > 10, y \in \mathbb{R}$$

$$8y + 12 - 6y + 1 > 10$$

$$8y - 6y + 1 + 12 > 10$$

$$2y + 13 > 10$$

$$2y > 10 - 13$$

$$2y > -3$$

$$y > \frac{-3}{2}$$

$$\left\{ y \mid y \in \mathbb{R}, y > \frac{-3}{2} \right\}$$

Q: 2(n)(or)

A: 2(n)(or)

$$B = \frac{3+n}{2}, \frac{4+y}{2} = (-1, 7)$$

$$\frac{3+n}{2} = \frac{-1}{1}, \frac{4+y}{2} = \frac{7}{1}$$

$$3+n = -2$$

$$4+y = 14$$

$$n = -2 - 3$$

$$y = 14 - 4$$

$$n = -5$$

$$y = 10$$

Q: 2(ni)(or)

A: 2(ni)(or)

$$\frac{n+(-3)}{2}, \frac{4+y}{2} = 2, 3$$

$$\frac{n-3}{2} = 2, \frac{4+y}{2} = 3$$

$$n-3 = 4$$

$$4+y = 6$$

$$n = 4 + 3$$

$$y = 6 - 4$$

$$n = 7$$

$$y = 2$$

$$\begin{aligned} & \text{Q: 3(i)} \\ & \text{A: 3(i)} \\ \log 0.0036 &= \frac{36}{10000} \end{aligned}$$

$$\frac{2^2 \times 3^2}{2^4 \times 5^4}$$

$$= \frac{2^2 \times 3^2}{2^2 \times 2^2 \times 5^4}$$

$$\log = \frac{3^2}{2^2 \times 5^4}$$

$$\begin{aligned} \log 3^2 &= \log 2^2 - \log 5^4 \\ 2 \log 3 &= 2 \log 2 - 4 \log 5 \\ 2(0.4771) &= 2(0.3010) - 4(0.6990) \\ 0.9542 &= 0.6020 - 2.7960 \\ &= -2.4438 \end{aligned}$$

This is negative with

$$\begin{aligned} &= -2.4438 + 3 - 3 = 0.5562 - 3 \\ &= -2.4438 \end{aligned}$$

Q: 3(ii)

A: 3(ii)

$$x^3 p^2 - 8y^3 p^2 - 4x^2 q^2 + 32y^3 p q$$

$$p^2(x^3 - 8y^3) - 4q^2(x^2 - 8y^3)$$

$$(x^3 - 8y^3)(p^2 - 4q^2)$$

$$\text{factor } x^3 - 8y^3 = (x)^3 - (2y)^3 =$$

$$(x-2y)(x^2 + (x)(2y) + (2y)^2) \\ (x-2y)(x^2 + 2xy + 4y^2)$$

$$\text{factor } p^2 - 4q^2 = (p-2q)(p+2q)$$

$$(p^2 - (2q)^2) \\ (p-2q)(p+2q)$$

$$\text{Result :- } (x-2y)(x^2 + 2xy + 4y^2)(p-2q)(p+2q)$$

Q: 3(iii)(or)

A: 3(iii)(or)

$$\textcircled{a} \left[\frac{\sqrt{5x-4} - 4}{10} \right] = -1$$

$$\sqrt{5x-4} - 4 = -10$$

$$\sqrt{5x-4} = -10 + 4$$

$$\sqrt{5x-4} = -6$$

$$\{ \emptyset \}$$

$$1.3n - 0.3n = -1.5 + 0.2$$

$$1.0n = -1.3$$

$$n = -1.3$$

$$\text{Q: } 3(\text{iv}) \text{ (or) } =$$

$$\text{A: } 3(\text{iv}) \text{ (or) } =$$

∴ midpoint of AB :-

$$\frac{3-4}{2}, \frac{7+0}{2}$$

$$\frac{-1}{2}, \frac{7}{2}$$

$$-0.5, 3.5$$

∴ midpoint of AC :-

$$\frac{3+4}{2}, \frac{7+0}{2}$$

$$\frac{7}{2}, \frac{7}{2}$$

$$3.5, 3.5$$

$$AC = \sqrt{(3.5 - (-0.5))^2 + (3.5 - 3.5)^2}$$

$$= \sqrt{(4)^2 + (0)^2}$$

$$= \sqrt{16 + 0}$$

$$4$$

$$d_{BC} = \sqrt{(4 - (-4))^2 + (0)^2}$$

$$= \sqrt{8^2 + 0}$$
$$= \sqrt{\frac{64}{8}}$$

MCQs:-

1. B

2. C

3. A

4. B

5. A

6. B

7. C

8. B

9. C

10. B

11. C

12. C